

POC / MVP • AI Voice Automation • BGV Operations

AI Voice Agent for BGV Insufficiency Resolution

A voice-driven solution that automates candidate follow-up for background verification insufficiencies, reduces analyst effort, and accelerates case resolution through structured conversation and record updates.

Version: 1.0 **Prepared for:** Akash / Samatha **Prepared by:** Shravya, Inayat, Saketh

Stack: Dograh AI + Twilio + FastMCP + SQLite + Python

[View related PDF](#)

1. Design Planning Summary

Background

BGV teams often face incomplete records, invalid references, missing documents, and unresolved verification issues. Analysts spend significant time manually contacting candidates, tracking responses, and updating records.

Proposed Solution

The solution introduces an AI voice agent that automatically calls candidates when an insufficiency is detected, collects required information, updates records, and routes exceptions when needed.

2. Business Value

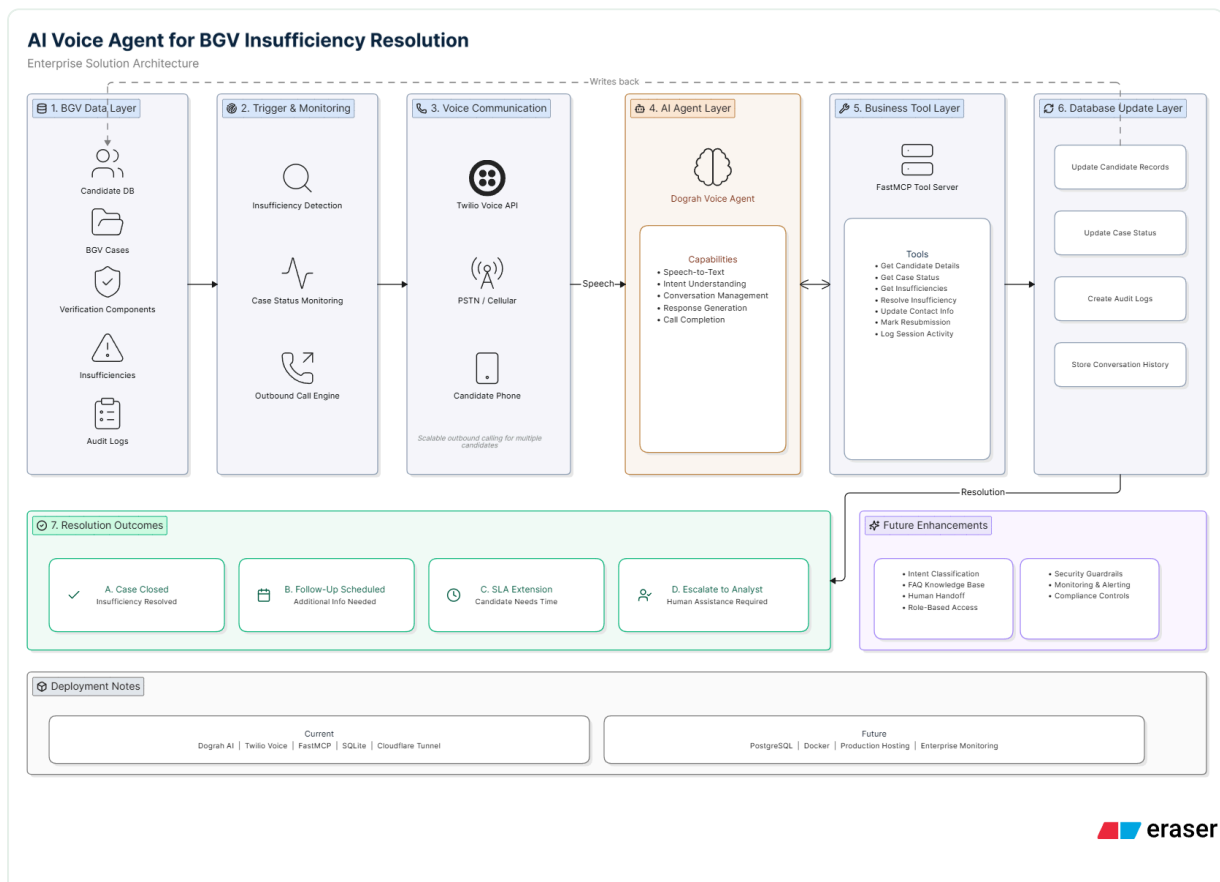
- › Reduces analyst effort spent on repetitive candidate follow-ups.
- › Improves insufficiency resolution turnaround time.
- › Prevents onboarding delays and supports SLA compliance.

- › Creates a standard, auditable process for every candidate interaction.

3. Functional Overview

- › Candidate case is reviewed and an insufficiency is identified.
- › The voice agent is triggered and a call is initiated through Twilio.
- › The agent collects missing details through a guided conversation.
- › Relevant records are updated and the interaction is logged.
- › The case is resolved or escalated to an analyst.

4. Architecture at a Glance



5. Current Progress Status

Completed

Validated

In Progress

- Database integration and candidate retrieval
- Insufficiency detection and automated call triggering
- Twilio voice integration and Dograh AI orchestration
- Read/write operations, session logging, and audit trail
- Contact update and escalation workflows

6. Key Challenges

- Network restrictions due to VPN and firewall limitations.
- Production deployment readiness needs security hardening and guardrails.
- Additional work is required in monitoring, access control, FAQ support, and intent classification.

7. Cost Estimate

Scenario	Total Cost (USD)	Total Cost (INR)
Call usage only	\$9.75	₹839
With 1 Twilio number	\$11.25	₹968
With 10 Twilio numbers	\$24.75	₹2,129

8. Next Steps

- > Implement security review, guardrails, and compliance checks.
- > Enhance agent intelligence with FAQ support and human escalation workflows.
- > Refine deployment architecture, monitoring, and scaling readiness.
- > Validate larger-scale usage and alternate voice-agent options.

The AI Voice Agent for BGV Insufficiency Resolution demonstrates a working end-to-end proof of concept and provides a scalable foundation for future enterprise deployment.